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A thermodynamic description of the Mg–Pr–Y system

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Abstract

The thermodynamic modeling and optimization of the Mg–Pr, Pr–Y binary systems and Mg–Pr–Y ternary system were critically carried out by means of the CALPHAD (CALculation of PHAse Diagram) technique. The solution phases (liquid, body-centered cubic, hexagonal close-packed and double hexagonal close-packed) were modeled with the Redlich–Kister equation. The Compound Energy Model has been used to describe the thermodynamic functions of the intermetallic compounds in these systems. The compounds Mg_2Y, $Mg_{12}Pr$, $Mg_{41}Pr_5$, Mg_3Pr, Mg_2Pr, Mg_5M and $Mg_{24}Y_5$ in the Mg–Pr–Y system were treated as the formulae $(Mg, Y)_2(Mg, Pr, Y)$, $Mg_{12}(Pr, Y)$, $Mg_{41}(Pr, Y)_5$, $Mg_3(Mg, Pr, Y)$, $Mg_2(Pr, Y)$, $Mg_5(Pr, Y)$ and $Mg_{24}(Mg, Pr, Y)_4Y$, respectively. A two-sublattice model $(Mg, Pr, Y)_{0.5}(Mg, Pr, Y)_{0.5}$ was applied to describe the compound MgM formed by MgPr and MgY in order to cope with the order–disorder transition between body-centered cubic solution (A2) and MgM with CsCl-type structure (B2) in the Mg–Pr–Y system. A set of self-consistent thermodynamic descriptions of the Mg–Pr–Y system was obtained. The projection of the liquidus surfaces for the Mg–Pr–Y system and the complete reaction scheme for the Mg–Pr–Y system were predicted.
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Keywords: Mg–Pr–Y system; Thermodynamic modeling; CALPHAD technique

1. Introduction

Magnesium-based alloys are considered to be the most promising materials for hydrogen storage because of their hydrogen storage capacity, their light weight nature, the abundance of magnesium in the earth's crust, and their low-cost compared with alternative systems. Yttrium and other RE-additives are of interest because they enhance high-temperature properties and improve casting characteristics. The Mg–Y alloys show higher creep resistance, better corrosion resistance and a considerable age hardening response. They have good strength properties at room temperature as well as at high temperatures.

In order to obtain a thermodynamic description of a multi-component system, a thermodynamic description of each lower-order system is necessary. Mg–Y [1,2] and Mg–Pr [3] systems have been optimized. Mg–Pr system was re-optimized in order to coincide exactly with the ternary phase equilibria in the Mg–Pr–Y system. Meanwhile, Pr–Y system also was assessed while optimizing the ternary Mg–Pr–Y system.

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2. Experimental information

2.1. Mg–Pr system

The Mg–Pr system was first investigated by Canneri [4] using thermal analysis and metallographic methods and three intermetallic compounds Mg_3Pr, MgPr and $MgPr_4$ were reported. The purity of the starting materials was not specified. Later, the Mg–Pr phase diagram was re-investigated by Rolla et al. [5], Park and Wyman [6] and Bochvar [7], respectively. However, there is no general agreement concerning the stoichiometry of the compounds involved in the Mg-rich eutectic and the coordinates of the eutectic point [4,6,7]. Joseph and Gschneidner [8] determined the solid solubility of Mg in dhcp(Pr) by lattice parameter measurements. Vogel and Heumann [9] and Iandelli [10] confirmed the existence of Mg_2Pr and determined the crystal structure. Evdokimenko and Kripyakevich [11] determined the crystal structure of a Mg-rich compound and reported its stoichiometry to be $Mg_{12}Pr$. Kripyakevich and Evdokimenko [12] reported a second Mg-rich compound "$\sim Mg_8Pr$" with the $Mg_{41}Ce_5$-type structure, however, Johnson and Smith [13] considered

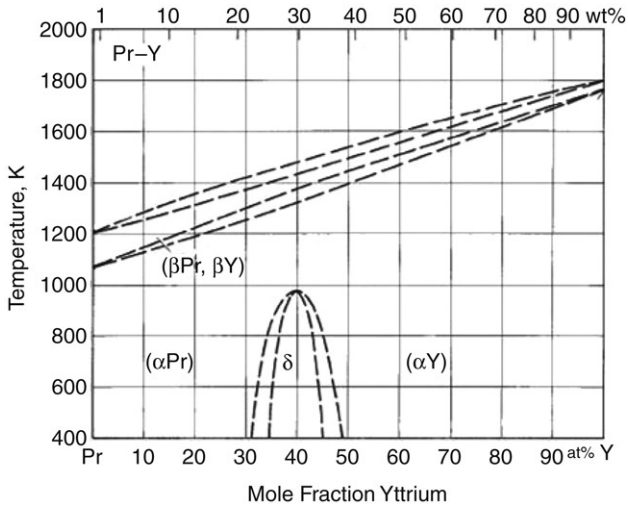

Fig. 1. The Pr–Y phase diagram assessed by Massalski et al. [22].

that it should be $Mg_{41}Pr_5$. Based on the above experimental information, the Mg–Pr system was assessed by Nayeb-Hashemi and Clark [14], five intermetallic compounds $Mg_{12}Pr$, $Mg_{41}Pr_5$, Mg_3Pr, Mg_2Pr and MgPr were presented, in which Mg_3Pr melts congruently, and the others were formed by peritectic reactions. Recently, Saccone et al. [15,16] re-investigated the Mg–Pr phase diagram and found that the compound MgPr melts congruently, and is not formed by a peritectic reaction as reported in previous works [6,7,14]. Saccone et al. [15] measured the Mg-rich region and Pr-rich region by Smith thermal analysis again, and modified the temperatures of some invariant reactions.

Canneri and Rossi [17] determined the enthalpies of formation of MgPr and Mg_3Pr by the acid-solution method. In the present work, the enthalpies of formation of the compounds $Mg_{12}Pr$, $Mg_{41}Pr_5$, Mg_3Pr and MgPr were predicted using the revised Miedema model of Niessen et al. [18], in kJ/mol of atoms: $Mg_{12}Pr$, −2.72; $Mg_{41}Pr_5$, −3.84; Mg_3Pr, −8.74 and MgPr, −12.2.

2.2. Pr–Y system

An experimentally determined phase diagram is not known. On the basis of information given by Harris et al. [19] and Gschneidner et al. [20], Moffatt [21] has sketched a phase diagram, which has been redrawn by Massalski et al. [22] and which has been used as a source of information for constructing Fig. 1. In the sketched phase diagram, the solid solution phases dhcp(Pr) and hcp(Y) form a homogeneous solid solution, which is not in accordance with the phase diagram regulation.

2.3. Mg–Pr–Y system

The isothermal sections of the Mg-rich side in the Mg–Pr–Y system at 773 K were determined by Giovannini et al. [23]. The compounds MgPr and MgY have the same structure of B2, and form solid solution over the complete composition range. The ternary compounds λ with Cu_2Mg structure was observed. The phase based on the Mg_2Y shows a homogeneity field, which already was observed by Flandorfer et al. [24]. For the compound Mg_3Pr, due to the addition of Y atom, a large solution range has been observed for this Mg_3Pr-based phase of cubic BiF_3 type. The ternary compound Mg_5M was found in the Mg-rich region. In addition, the Mg-rich binary phases $Mg_{24}Y_5$, $Mg_{12}Pr$ and $Mg_{41}Pr_5$ show some ternary solubilities.

3. Thermodynamic model

The Gibbs energy of all the solution phases (liquid, hcp, dhcp, bcc) in the Mg–Pr–Y system is described by the substitution solution model in the present work. The binary compounds Mg_3Pr in the Mg–Pr system and Pr_2Y in the Pr–Y system have a homogeneity range, and are treated as the formula $Mg_3(Mg, Pr)$ and $(Pr, Y)_2(Pr, Y)$, respectively. The compound MgPr has a B2 structure. In order to reflect the relationship between the disordered A2 phase and the ordered B2 MgPr phase, the same formula of Gibbs energy is adopted for the two phases. The other binary compounds in these two binary systems are treated as stoichiometric compounds.

3.1. Unary phases

The Gibbs energy function $G_i^{\phi}(T) = G_i^{0,\phi}(T) - H_i^{SER}$ (298.15 K) for the element i (i = Mg, Pr, Y) in the phase ϕ (ϕ = liquid, body-centered cubic (bcc), double hexagonal close-packed (dhcp) or hexagonal close-packed (hcp)) is described by an equation of the following form:

$$G_i^{\phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where H_i^{SER} (298.15 K) is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, hcp for Mg and Y, dhcp for Pr. The Gibbs energy of the element i, $G_i^{\phi}(T)$, in its SER state, is denoted by $GHSER_i$, i.e.,

$$GHSER_i = {}^0G_i^{SER}(T) - H_i^{SER}(298.15). \quad (2)$$

In the present work, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [25].

3.2. Solution phases

In the Mg–Pr–Y system, there are four solution phases: liquid, bcc, dhcp and hcp. Their Gibbs energies are described by the following expression:

$$G_m^{\phi} = x_{Mg}G_{Mg}^{\phi}(T) + x_{Pr}G_{Pr}^{\phi}(T) + x_Y G_Y^{\phi}(T) + RT(x_{Mg}\ln x_{Mg} + x_{Pr}\ln x_{Pr} + x_Y \ln x_Y) + {}^EG_m^{\phi} \quad (3)$$

where x_{Mg}, x_{Pr} and x_Y are mole fractions of the pure elements Mg, Pr and Y, respectively; ${}^EG_m^{\phi}$ is the excess Gibbs energy, expressed by the Redlich–Kister polynomial [26],

$${}^EG_m^{\phi} = x_{Mg}x_{Pr}\sum_i {}^iL_{Mg,Pr}^{\phi}(x_{Mg} - x_{Pr}) + x_{Pr}x_Y\sum_i {}^iL_{Pr,Y}^{\phi}(x_{Pr} - x_Y)$$

Table 1
Thermodynamic parameters in the Mg–Pr–Y system[a]

Phase	Thermodynamic parameters
Liquid: $(Mg, Pr, Y)_1$	$G(\text{liquid, Pr}) =$
	298–1068 $+3848.961 - 29.099465T - 4.7344931 * T * \ln T$
	$-.035119723T^2 + 5.427467 * 10^{-6} * T^3 - 207\,406 * T^{-1}$
	1068–3800 $-10\,539.574 + 219.508805 * T - 42.9697 * T * \ln T$
	$G(\text{liquid, Mg}) =$
	298–923 $+8202.243 - 8.83693T + \text{GHSER}_{\text{Mg}} - 8.0176 * 10^{-20}T^7$
	923–3000 $-5439.869 + 195.324057T - 34.3088T \ln T$
	$G(\text{liquid, Y}) =$
	100–1000 $+2098.50738 + 119.41873T - 24.6467508T \ln T$
	$-0.00347023463T^2 - 8.12981167 * 10^{-7}T^3$
	$+23\,713.7332T^{-1}$
	1000–1795 $+7386.44846 + 19.4520171T - 9.0681627T \ln T$
	$-0.0189533369T^2 + 1.7595327 * 10^{-6}T^3$
	1795–3700 $-12\,976.5957 + 257.400783T - 43.0952T \ln T$
	${}^0L^{\text{liq.}}_{\text{Mg,Pr}} = -37\,966.2 + 10.4036T$
	${}^1L^{\text{liq.}}_{\text{Mg,Pr}} = -32\,661.1 + 20.5106T$
	${}^0L^{\text{liq.}}_{\text{Mg,Y}} = -41\,194.9 + 17.5452T$
	${}^1L^{\text{liq.}}_{\text{Mg,Y}} = -15\,779.0 + 4.7064T$
	$\text{GHSER}_{\text{Pr}} =$
	298–500 $-18\,803.379 + 356.587384T - 68.9176 * T * \ln T$
	$+0.72929T^2 - 2.5184333 * 10^{-5} * T^3 + 507\,385 * T^{-1}$
	500–800 $-7246.848 + 82.427384 * T - 22.8909 * T * \ln T$
	$-0.00497126T^2 - 1.22951 * 10^{-6} * T^3 + 507\,385 * T^{-1}$
	800–1068 $+95\,411.023 - 1073.55111T + 146.764 * T * \ln T$
	$-0.1288205T^2 + 1.5592233 * 10^{-5} * T^3 - 11\,588\,800 * T^{-1}$
	1068–1204 $-481\,663.131 + 4234.33311T - 606.120311 * T * \ln T$
	$+0.305181506T^2 + 3.0994702 * 10^{-5} * T^3 - 70\,926\,840 * T^{-1}$
	1204–3800 $-20\,014.678 + 227.685155T - 42.9697 * T * \ln T$
	$\text{GHSER}_{\text{Mg}} =$
	298–923 $-8367.34 + 143.675547T - 26.1849782T \ln T$
	$+4.858 * 10^{-4}T^2 - 1.393669 * 10^{-6}T^3 + 78\,950T^{-1}$
	923–3000 $-14\,130.185 + 204.716215T - 34.3088T \ln T$
	$+1.038192 * 10^{28}T^{-9}$
	$\text{GHSER}_{\text{Y}} =$
	100–1000 $-8011.09379 + 128.572856T - 25.6656992T \ln T$
	$-0.00175716414T^2 - 4.17561786 * 10^{-7}T^3$
	$+26\,911.509T^{-1}$
	1000–1795 $-7179.74574 + 114.497104T - 23.4941827T \ln T$
	$-0.0038211802T^2 - 8.2534534 * 10^{-8}T^3$
	1795–3700 $-67\,480.7761 + 382.124727T - 56.9527111T \ln T$
	$+0.00231774379T^2 - 7.22513088 * 10^{-8}T^3$
	$+18\,077\,162.6T^{-1}$
bcc (disordered part of B2): $(Mg, Pr, Y)_1$	$G(\text{bcc, Pr; 0}) =$
	298–1068 $-2863.651 + 28.274853 * T - 13.7470527 * T * \ln T$
	$-0.02284377 * T^2 + 3.542468 * 10^{-6} * T^3 - 87\,486T^{-1}$
	1068–1204 $-11\,985.919 + 188.657121T - 38.451T \ln T$
	1204–3800 $+953.224 + 10.826821T - 26.6824313T \ln T$
	$-0.004106833T^2 + 1.76214 * 10^{-7}T^3 - 2\,473\,024T^{-1}$
	$G(\text{bcc, Mg; 0}) =$
	298–3000 $+3100 - 2.1T + \text{GHSER}_{\text{Mg}}$
	$G(\text{bcc, Y; 0}) =$
	100–1000 $-833.658863 + 123.667346T - 25.5832578T \ln T$
	$-0.00237175965T^2 + 9.10372497 * 10^{-9}T^3$
	$+27\,340.0687T^{-1}$
	1000–1795 $-1297.79829 + 134.528352T - 27.3038477T \ln T$
	$-5.41757644 * 10^{-4}T^2 - 3.05012175 * 10^{-7}T^3$
	1795–3700 $+15\,389.4975 + 0.981325399T - 8.88296647T \ln T$
	$-0.00904576576T^2 + 4.02944768 * 10^{-7}T^3$
	$-2\,542\,575.96T^{-1}$
	${}^0L^{\text{bcc}}_{\text{Mg,Y}} = -38\,394.5 + 15.7048T$

(*continued on next page*)

Table 1 (*continued*)

Phase	Thermodynamic parameters
	${}^{1}L^{\mathrm{bcc}}_{\mathrm{Mg,Y}} = +3112.9 - 5.0608T$
	${}^{0}L^{\mathrm{bcc.}}_{\mathrm{Mg,Pr}} = -33\,798.4 + 6.2198T$
	${}^{1}L^{\mathrm{bcc.}}_{\mathrm{Mg,Pr}} = -14\,006.9$
	${}^{2}L^{\mathrm{bcc.}}_{\mathrm{Mg,Pr}} = -5977.8$
hcp: $(\mathrm{Mg, Pr, Y})_1$	$G(\mathrm{hcp, Pr}; 0) = +\mathrm{GHSER_{Pr}} + 303.4$
	$G(\mathrm{hcp, Mg}; 0) = +\mathrm{GHSER_{Mg}}$
	$G(\mathrm{hcp, Y}; 0) = +\mathrm{GHSER_{Y}}$
	${}^{0}L^{\mathrm{hcp.}}_{\mathrm{Mg,Pr}} = -1000.0$
	${}^{0}L^{\mathrm{hcp}}_{\mathrm{Pr,Y}} = -5229.2 + 3.6752T$
	${}^{0}L^{\mathrm{hcp}}_{\mathrm{Mg,Y}} = -26\,612.8 + 13.9461T$
	${}^{1}L^{\mathrm{hcp}}_{\mathrm{Mg,Y}} = -2836.2$
dhcp: $(\mathrm{Mg, Pr, Y})_1$	$G(\mathrm{dhcp, Pr}; 0) = +\mathrm{GHSER_{Pr}}$
	$G(\mathrm{dhcp, Mg}; 0) = +\mathrm{GHSER_{Mg}} + 303.4$
	$G(\mathrm{dhcp, Y}; 0) = +\mathrm{GHSER_{Y}} + 303.4$
	${}^{0}L^{\mathrm{dhcp}}_{\mathrm{Mg,Pr}} = +21\,774.3 - 10.2124T$
	${}^{1}L^{\mathrm{dhcp}}_{\mathrm{Mg,Pr}} = +45\,373.2 - 14.5329T$
	${}^{0}L^{\mathrm{dhcp}}_{\mathrm{Pr,Y}} = -4727.4 + 3.6664T$
MgM (ordered part of B2):	$G^{\mathrm{MgM}}_{\mathrm{Pr:Mg}} = -23\,992.1 + 1.8180T$
$(\mathrm{Mg, Pr, Y})_{0.5}(\mathrm{Mg, Pr, Y})_{0.5}$	$G^{\mathrm{MgM}}_{\mathrm{Mg:Pr}} = -23\,992.1 + 1.8180T$
	$G^{\mathrm{MgM}}_{\mathrm{Mg:Y}} = -26\,294.8 + 1.4238T$
	$G^{\mathrm{MgM}}_{\mathrm{Y:Mg}} = -26\,294.8 + 1.4238T$
	${}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg:Mg,Pr}} = {}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg,Pr:Mg}} = -17\,630.4 - 9.4602T$
	${}^{1}L^{\mathrm{MgM}}_{\mathrm{Mg:Mg,Pr}} = {}^{1}L^{\mathrm{MgM}}_{\mathrm{Mg,Pr:Mg}} = -10\,133.8$
	${}^{0}L^{\mathrm{MgM}}_{\mathrm{Pr:Mg,Pr}} = {}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg,Pr:Pr}} = +8488.1$
	${}^{0}L^{\mathrm{MgM}}_{\mathrm{Y:Mg,Y}} = {}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Y}} = -20\,353.1$
	${}^{1}L^{\mathrm{MgM}}_{\mathrm{Y:Mg,Y}} = {}^{1}L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Y}} = +12\,647.3$
$\mathrm{Mg_2Pr} : \mathrm{Mg_2(Pr, Y)}$	$G^{\mathrm{Mg_2Pr}}_{\mathrm{Mg:Pr}} = 2\mathrm{GHSER_{Mg}} + \mathrm{GHSER_{Pr}} - 32\,085.7 + 5.3573T$
	$G^{\mathrm{Mg_2Pr}}_{\mathrm{Mg:Y}} = 2\mathrm{GHSER_{Mg}} + \mathrm{GHSER_{Y}} - 34\,890.5 + 6.6000T$
	${}^{0}L^{\mathrm{Mg_2Pr}}_{\mathrm{Mg:Pr,Y}} = +10\,000.0 - 22.0000T$
	${}^{1}L^{\mathrm{Mg_2Pr}}_{\mathrm{Mg:Pr,Y}} = +5000.0$
$\mathrm{Mg_2Y} : \mathrm{(Mg, Y)_2(Mg, Pr, Y)}$	$G^{\mathrm{Mg_2Y}}_{\mathrm{Mg:Pr}} = 2\mathrm{GHSER_{Mg}} + \mathrm{GHSER_{Pr}} - 33\,760.9 - 2.5370T$
	$G^{\mathrm{Mg_2Y}}_{\mathrm{Mg:Y}} = 2\mathrm{GHSER_{Mg}} + \mathrm{GHSER_{Y}} - 37\,147.9 + 6.0541T$
	$G^{\mathrm{Mg_2Y}}_{\mathrm{Y:Mg}} = \mathrm{GHSER_{Mg}} + 2\mathrm{GHSER_{Y}} + 67\,147.9 - 6.0541T$
	$G^{\mathrm{Mg_2Y}}_{\mathrm{Mg:Mg}} = 3\mathrm{GHSER_{Mg}} + 15\,000.0$
	$G^{\mathrm{Mg_2Y}}_{\mathrm{Y:Y}} = 3\mathrm{GHSER_{Y}} + 15\,000.0$
	${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Mg:Mg,Y}} = -21\,000.0 + 7.5T$
	${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Y:Mg,Y}} = -21\,000.0 + 7.5T$
	${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Mg,Y:Mg}} = +21\,000.0$
	${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Mg,Y:Y}} = +21\,000.0$
$\mathrm{Mg_3Pr} : \mathrm{Mg_3(Pr, Y)}$	$G^{\mathrm{Mg_3Pr}}_{\mathrm{Mg:Pr}} = 3\mathrm{GHSER_{Mg}} + \mathrm{GHSER_{Pr}} - 55\,568.2 + 7.9084T$
	$G^{\mathrm{Mg_3Pr}}_{\mathrm{Mg:Mg}} = 4\mathrm{GHSER_{Mg}} + 12\,400 - 8.4T$
	$G^{\mathrm{Mg_3Pr}}_{\mathrm{Mg:Mg,Pr}} = +11\,498.0$
	$G^{\mathrm{Mg_3Pr}}_{\mathrm{Mg:Y}} = 3\mathrm{GHSER_{Mg}} + \mathrm{GHSER_{Y}} - 34\,322.9 + 6.5953T$
	$G^{\mathrm{Mg_3Pr}}_{\mathrm{Mg:Pr,Y}} = -4000.0$
$\mathrm{Mg_{41}Pr_5} : \mathrm{Mg_{41}(Pr, Y)_5}$	$G^{\mathrm{Mg_{41}Pr_5}}_{\mathrm{Mg:Pr}} = 41\mathrm{GHSER_{Mg}} + 5\mathrm{GHSER_{Pr}} - 307\,733.4 + 58.4895T$
	$G^{\mathrm{Mg_{41}Pr_5}}_{\mathrm{Mg:Y}} = 41\mathrm{GHSER_{Mg}} + 5\mathrm{GHSER_{Y}} - 198\,945.6 + 7.2306T$

Table 1 (*continued*)

Phase	Thermodynamic parameters
$Mg_{12}Pr$: $Mg_{12}(Pr, Y)$	$G^{Mg_{12}Pr}_{Mg:Pr} = 12GHSER_{Mg} + GHSER_{Pr} - 62\,630.2 + 12.8287T$
	$G^{Mg_{12}Pr}_{Mg:Y} = 12GHSER_{Mg} + GHSER_{Y} - 38\,920.4 + 1.4461T$
$Mg_{24}Y_5$: $Mg_{24}(Mg, Pr, Y)_4Y$	$G^{Mg_{24}Y_5}_{Mg:Mg:Y} = 28GHSER_{Mg} + GHSER_{Y} - 5932.0$
	$G^{Mg_{24}Y_5}_{Mg:Y:Y} = 24GHSER_{Mg} + 5GHSER_{Y} - 202\,101.8 + 7.2306T$
	$G^{Mg_{24}Y_5}_{Mg:Pr:Y} = 24GHSER_{Mg} + 4GHSER_{Pr} + GHSER_{Y} - 200\,000.0 + 20.0000T$
	${}^0L^{Mg_{24}Y_5}_{Mg:Mg,Pr,Y:Y} = +10\,000.0$
Mg_5M : $Mg_5(Pr, Y)$	$G^{Mg_5M}_{Mg:Pr} = 5GHSER_{Mg} + GHSER_{Pr} - 57\,716.5 + 11.2134T$
	$G^{Mg_5M}_{Mg:Y} = 5GHSER_{Mg} + GHSER_{Y} - 39\,988.2 + 1.1170T$
	$L^{Mg_5M}_{Mg:Pr,Y} = -3814.1$
Pr_2Y : $(Pr, Y)_2(Pr, Y)$	$G^{Pr_2Y}_{Pr:Pr} = 3GHSER_{Pr} + 15\,000.0$
	$G^{Pr_2Y}_{Y:Y} = 3GHSER_{Y} + 15\,000.0$
	$G^{Pr_2Y}_{Pr:Y} = 2GHSER_{Pr} + GHSER_{Y} - 5451.5 - 10.0635T$
	$G^{Pr_2Y}_{Y:Pr} = GHSER_{Pr} + 2GHSER_{Y} + 35\,451.5 + 10.0635T$
	${}^0L^{Pr_2Y}_{Pr,Y:Pr} = {}^0L^{Pr_2Y}_{Pr,Y:Y} = +30\,250.1 + 30.7514T$

[a] In SI units (Joule, mole of the formula units and Kelvin).

Table 2
Invariant reactions in the Mg–Pr system

Reaction	Present work				Experimental data [15,16]			
	T(K)	x(Pr)			T(K)	x(Pr)		
liq. → bcc(Pr) + MgPr	1008	0.6097	0.6602	0.5249	1008	0.600	0.669	0.528
bcc(Pr) → MgPr + dhcp(Pr)	783	0.8097	0.5274	0.9279	783	0.805	0.508	0.895
liq. → MgPr	1038	0.4970	0.4970	–	1038	0.500	0.500	–
liq → MgPr + Mg_2Pr	998	0.3828	0.4618	–	998	0.405	0.460	–
liq. + Mg_3Pr → Mg_2Pr	1013	0.3479	0.2495	–	1013	0.391	0.250	–
liq. → Mg_3Pr	1063	0.2471	0.2471	–	1063	0.248	0.248	–
Mg_2Pr → Mg_3Pr + MgPr	943	0.2498	0.4688	–	943	0.250	0.475	–
liq. + Mg_3Pr → $Mg_{41}Pr_5$	848	0.0811	0.2194	–	848	0.084	0.180	–
liq. + $Mg_{41}Pr_5$ → $Mg_{12}Pr$	838	0.0676	–	–	838	0.074	–	–
liq. → hcp(Mg) + $Mg_{12}Pr$	837	0.0660	0.0006	–	833	0.050	0.001	–

$$+ x_{Mg}x_Y \sum_i {}^iL^{\phi}_{Mg,Y}(x_{Mg} - x_Y)^i$$
$$+ x_{Mg}x_{Pr}x_Y L^{\phi}_{Mg,Pr,Y} \quad (4)$$

where ${}^iL^{\phi}_{Mg,Pr}$, ${}^iL^{\phi}_{Pr,Y}$ and ${}^iL^{\phi}_{Mg,Y}$ are the interaction parameters between the elements Mg and Pr, Pr and Y, and Mg and Y, respectively. $L^{\phi}_{Mg,Pr,Y}$ is the ternary interaction parameter.

3.3. Intermetallic compounds

The binary intermetallic compound $Mg_{24}Y_5$ has an α (Mn) structure, Mg and Y atoms mix in the 8c position, while the 2a position is occupied by an Y atom according to the crystallographic data determined by Bonhomme and Yvon [27]. $Mg_{24}Y_5$ is treated as the formula $Mg_{24}(Mg, Y)_4Y$ according to Fabrichnaya et al. [1]. In the Mg–Pr–Y ternary system, the compound has a homogeneity range for Pr atom due to the addition of Pr. In order to agree with the compound in the Gd–Mg–Y system [2], it is treated as $Mg_{24}(Mg, Pr, Y)_4Y$. The Gibbs energy per mole of formula unit $Mg_{24}Y_5$ is given by the following expression:

$$\begin{aligned} G^{Mg_{24}Y_5} = {} & y_{Mg}G^{Mg_{24}Y_5}_{Mg:Mg:Y} + y_{Pr}G^{Mg_{24}Y_5}_{Mg:Pr:Y} + y_Y G^{Mg_{24}Y_5}_{Mg:Y:Y} \\ & + 4RT(y_{Mg}\ln y_{Mg} + y_{Pr}\ln y_{Pr} + y_Y \ln y_Y) \\ & + y_{Pr}y_{Mg}\sum_i {}^iL^{Mg_{24}Y_5}_{Mg:Pr,Mg:Y}(y_{Mg} - y_{Pr})^i \\ & + y_{Pr}y_Y \sum_i {}^iL^{Mg_{24}Y_5}_{Mg:Pr,Y:Y}(y_{Pr} - y_Y)^i \\ & + y_{Mg}y_Y \sum_i {}^iL^{Mg_{24}Y_5}_{Mg:Mg,Y:Y}(y_{Mg} - y_Y)^i \\ & + y_{Mg}y_{Pr}y_Y L^{Mg_{24}Y_5}_{Mg:Mg,Pr,Y:Y} \end{aligned} \quad (5)$$

where y_{Mg}, y_{Pr} and y_Y are the site fractions of Mg, Pr or Y on the second sublattice; three parameters denoted $G^{Mg_{24}Y_5}_{Mg:Mg:Y}$, $G^{Mg_{24}Y_5}_{Mg:Pr:Y}$ and $G^{Mg_{24}Y_5}_{Mg:Y:Y}$ represent the Gibbs energies of the compound $Mg_{24}Y_5$ when the second sublattice is occupied by only one element Mg, Pr or Y ($y_{Mg} = 1$, $y_{Pr} = 1$ or $y_Y = 1$), respectively, which are relative to the enthalpies

Table 3
Invariant reactions in the Pr–Y system

Reaction	Present work				Experimental data [22]			
	T(K)	x(Y)			T(K)	x(Y)		
dhcp+hcp → Pr_2Y	908	0.3256	0.3324	0.3117	903	–	–	–
bcc + hcp → dhcp	1277	0.2796	0.3309	0.3150	–	–	–	–

of pure hcp for Mg and Y and dhcp for Pr in their SER state; ${}^{i}L^{Mg_{24}Y_5}_{Mg:Mg,Y:Y}$, ${}^{i}L^{Mg_{24}Y_5}_{Mg:Mg,Pr:Y}$, ${}^{i}L^{Mg_{24}Y_5}_{Mg:Pr,Y:Y}$ represent the ith interaction parameters between the elements Mg and Y, Mg and Pr or Pr and Y on the second sublattice, respectively. The parameter ${}^{i}L^{Mg_{24}Y_5}_{Mg:Mg,Pr,Y:Y}$ represents the ith interaction parameter among the elements Mg, Pr and Y.

The intermetallic compound Mg_2Y has a $MgZn_2$ structure. In the binary Mg–Y system, it has the homogeneity range 66.7–74 at.% Mg [24] and is treated as the formula $(Mg, Y)_2(Mg, Y)$ according to the optimization results [1, 2]. In the Mg–Pr–Y ternary system, this compound shows a homogeneity field for Pr atom. In order to agree with the compound in the Gd–Mg–Y system [2], it is treated as $(Mg, Y)_2(Mg, Pr, Y)$. The Gibbs energy per mole of formula unit Mg_2Y is given by the following expression:

$$\begin{aligned}
G^{Mg_2Y} = {} & y'_{Mg}y''_{Mg}G^{Mg_2Y}_{Mg:Mg} + y'_{Mg}y''_{Pr}G^{Mg_2Y}_{Mg:Pr} \\
& + y'_{Mg}y''_{Y}G^{Mg_2Y}_{Mg:Y} + y'_{Y}y''_{Mg}G^{Mg_2Y}_{Y:Mg} + y'_{Y}y''_{Pr}G^{Mg_2Y}_{Y:Pr} \\
& + y'_{Y}y''_{Y}G^{Mg_2Y}_{Y:Y} + 2RT(y'_{Mg}\ln y'_{Mg} + y'_{Y}\ln y'_{Y}) \\
& + RT(y''_{Mg}\ln y''_{Mg} + y''_{Pr}\ln y''_{Pr} + y''_{Y}\ln y''_{Y}) \\
& + y'_{Mg}\Big[y''_{Pr}y''_{Mg}\sum {}^{i}L^{Mg_2Y}_{Mg:Mg,Pr}(y''_{Mg} - y''_{Pr})^{i} \\
& + y''_{Mg}y''_{Y}\sum {}^{i}L^{Mg_2Y}_{Mg:Mg,Y}(y''_{Mg} - y''_{Y})^{i} \\
& + y''_{Pr}y''_{Y}\sum {}^{i}L^{Mg_2Y}_{Mg:Pr,Y}(y''_{Pr} - y''_{Y})^{i} \\
& + y''_{Mg}y''_{Pr}y''_{Y}L^{Mg_2Y}_{Mg:Mg,Pr,Y}\Big] \\
& + y'_{Y}\Big[y''_{Pr}y''_{Mg}\sum {}^{i}L^{Mg_2Y}_{Y:Pr,Mg}(y''_{Mg} - y''_{Pr})^{i} \\
& + y''_{Mg}y''_{Y}\sum {}^{i}L^{Mg_2Y}_{Y:Mg,Y}(y''_{Mg} - y''_{Y})^{i} \\
& + y''_{Pr}y''_{Y}\sum {}^{i}L^{Mg_2Y}_{Y:Pr,Y}(y''_{Pr} - y''_{Y})^{i} \\
& + y''_{Mg}y''_{Pr}y''_{Y}L^{Mg_2Y}_{Y:Mg,Pr,Y}\Big] \\
& + y''_{Pr}\Big[y'_{Mg}y'_{Y}\sum {}^{i}L^{Mg_2Y}_{Mg,Y:Pr}(y'_{Mg} - y'_{Y})^{i}\Big] \\
& + y''_{Mg}\Big[y'_{Mg}y'_{Y}\sum {}^{i}L^{Mg_2Y}_{Mg,Y:Mg}(y'_{Mg} - y'_{Y})^{i}\Big] \\
& + y''_{Y}\Big[y'_{Mg}y'_{Y}\sum {}^{i}L^{Mg_2Y}_{Mg,Y:Y}(y'_{Mg} - y'_{Y})^{i}\Big].
\end{aligned} \tag{6}$$

The intermetallic compound MgM has a B2 structure and is formed over the complete composition range by the binary compounds MgPr and MgY. In order to reflect the structural relationship between the compound MgM (ordered, bcc-B2) and the bcc terminal solution phase (disordered, bcc-A2), in the present work, the two phases are modeled using a single formalism represented by $(Mg, Pr, Y)_{0.5}(Mg, Pr, Y)_{0.5}$. The Gibbs energy per mole of formula unit B2 MgM is given by the following expression:

$$G^{MgY}_m = G^{dis}_m(x_i) + G^{ord}_m(y'_i, y''_i) - G^{ord}_m(x_i) \tag{7}$$

where the mole fraction x_i and the site fractions y'_i, y''_i are related according to $x_i = 0.5y'_i + 0.5y''_i$; the first term $G^{dis}_m(x_i)$ is molar Gibbs energy of the disordered bcc phase and the expression is the same as Eq. (5); $G^{ord}_m(y'_i, y''_i) - G^{ord}_m(x_i)$ is the ordering contribution to the Gibbs energy. $G^{ord}_m(y'_i, y''_i)$ is calculated using Eq. (8) below, then $G^{ord}_m(x_i)$ is calculated using the same model parameters but replacing y'_i and y''_i by the mole fraction x_i in Eq. (8). When the phase becomes disordered, i.e. $y'_i = y''_i = x_i$, the contributions from the ordering, $G^{ord}_m(y'_i, y''_i) - G^{ord}_m(x_i)$, will be cancelled and only the first term $G^{dis}_m(x_i)$ gives a contribution to the Gibbs energy.

Table 4
Predicted invariant reactions in the Mg–Pr–Y system

Reaction	Type	T(K)	Composition in liquid		
			x(Mg)	x(Pr)	x(Y)
liq. → MgM + bcc	e1	1228			
liq. → MgM + Mg_2Pr	e2	1049			
liq. → Mg_3PrMg_2Pr	e3	1021			
liq. + MgM → Mg_2Pr + Mg_2Y	U_7	1011	0.7204	0.0578	0.2218
liq. + Mg_2Pr → Mg_3Pr + Mg_2Y	U_6	921	0.7934	0.0739	0.1327
liq → Mg_3Pr + Mg_5M	e4	879			
liq. + Mg_2Y → Mg_3Pr + Mg_5M	U_5	878	0.8294	0.0513	0.1193
liq. + Mg_2Y → $Mg_{24}Y_5$ + Mg_5M	U_4	875	0.8517	0.0176	0.1307
liq. + Mg_3Pr → $Mg_{41}Pr_5$ + Mg_5M	U_3	839	0.9030	0.0661	0.0309
liq. + $Mg_{12}Pr$ → $Mg_{41}Pr_5$ + hcp	U_2	834	0.9304	0.0624	0.0072
liq. + $Mg_{24}Y_5$ → hcp + Mg_5M	U_1	827	0.9083	0.0176	0.0741
liq → $Mg_{41}Pr_5$ + hcp + Mg_5M	E_1	824	0.9126	0.0385	0.0489
bcc + hcp → MgM + dhcp	U_8	825	–	–	–
dhcp + hcp → MgM + Pr_2Y	U_9	696	–	–	–

The intermetallic compound λ with Cu_2Mg structure has the same structure as the intermetallic compound Mg_2Pr. Giovannini et al. [23] think that the homogeneity field of the phase probably extends at high temperature, towards the Mg–Pr edge, possibly reaching the binary high temperature compound Mg_2Pr. In the present work, the compounds λ and Mg_2Pr is treated as an intermetallic compound and expressed as Mg_2(Pr, Y). The compounds $Mg_{12}Pr$, $Mg_{41}Pr_5$ have small ternary solubilities and are treated as Mg_m(Pr, Y). The ternary compound Mg_5M found in the Mg-rich

$$
\begin{aligned}
G_m^{\mathrm{ord}} = {} & y'_{\mathrm{Pr}} y''_{\mathrm{Pr}} G^{\mathrm{MgM}}_{\mathrm{Pr:Pr}} + y'_{\mathrm{Pr}} y''_{\mathrm{Mg}} G^{\mathrm{MgM}}_{\mathrm{Pr:Mg}} + y'_{\mathrm{Pr}} y''_{\mathrm{Y}} G^{\mathrm{MgM}}_{\mathrm{Pr:Y}} \\
& + y'_{\mathrm{Mg}} y''_{\mathrm{Pr}} G^{\mathrm{MgM}}_{\mathrm{Mg:Pr}} + y'_{\mathrm{Mg}} y''_{\mathrm{Mg}} G^{\mathrm{MgM}}_{\mathrm{Mg:Mg}} + y'_{\mathrm{Mg}} y''_{\mathrm{Y}} G^{\mathrm{MgM}}_{\mathrm{Mg:Y}} \\
& + y'_{\mathrm{Y}} y''_{\mathrm{Pr}} G^{\mathrm{MgM}}_{\mathrm{Y:Pr}} + y'_{\mathrm{Y}} y''_{\mathrm{Mg}} G^{\mathrm{MgM}}_{\mathrm{Y:Mg}} + y'_{\mathrm{Y}} y''_{\mathrm{Y}} G^{\mathrm{MgM}}_{\mathrm{Y:Y}} \\
& + y'_{\mathrm{Pr}} \Big[y''_{\mathrm{Pr}} y''_{\mathrm{Mg}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr:Pr,Mg}} (y''_{\mathrm{Mg}} - y''_{\mathrm{Pr}})^i \\
& + y''_{\mathrm{Mg}} y''_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr:Mg,Y}} (y''_{\mathrm{Mg}} - y''_{\mathrm{Y}})^i \\
& + y''_{\mathrm{Pr}} y''_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr:Pr,Y}} (y''_{\mathrm{Pr}} - y''_{\mathrm{Y}})^i \\
& + \ y''_{\mathrm{Pr}} y''_{\mathrm{Mg}} y''_{\mathrm{Y}} L^{\mathrm{MgM}}_{\mathrm{Pr:Pr,Mg,Y}} \Big] \\
& + y'_{\mathrm{Mg}} \Big[y''_{\mathrm{Pr}} y''_{\mathrm{Mg}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Mg:Pr,Mg}} (y''_{\mathrm{Mg}} - y''_{\mathrm{Pr}})^i \\
& + y''_{\mathrm{Mg}} y''_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Mg:Mg,Y}} (y''_{\mathrm{Mg}} - y''_{\mathrm{Y}})^i \\
& + y''_{\mathrm{Pr}} y''_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Mg:Pr,Y}} (y''_{\mathrm{Pr}} - y''_{\mathrm{Y}})^i \\
& + \ y''_{\mathrm{Pr}} y''_{\mathrm{Mg}} y''_{\mathrm{Y}} L^{\mathrm{MgM}}_{\mathrm{Mg:Pr,Mg,Y}} \Big] \\
& + y'_{\mathrm{Y}} \Big[y''_{\mathrm{Pr}} y''_{\mathrm{Mg}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Y:Pr,Mg}} (y''_{\mathrm{Mg}} - y''_{\mathrm{Pr}})^i \\
& + y''_{\mathrm{Mg}} y''_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Y:Mg,Y}} (y''_{\mathrm{Mg}} - y''_{\mathrm{Y}})^i \\
& + y''_{\mathrm{Pr}} y''_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Y:Pr,Y}} (y''_{\mathrm{Pr}} - y''_{\mathrm{Y}})^i \\
& + \ y''_{\mathrm{Pr}} y''_{\mathrm{Mg}} y''_{\mathrm{Y}} L^{\mathrm{MgM}}_{\mathrm{Y:Pr,Mg,Y}} \Big] \\
& + y''_{\mathrm{Pr}} \Big[y'_{\mathrm{Pr}} y'_{\mathrm{Mg}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr,Mg:Pr}} (y'_{\mathrm{Mg}} - y'_{\mathrm{Pr}})^i \\
& + y'_{\mathrm{Mg}} y'_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Pr}} (y'_{\mathrm{Mg}} - y'_{\mathrm{Y}})^i \\
& + y'_{\mathrm{Pr}} y'_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr,Y:Pr}} (y'_{\mathrm{Pr}} - y'_{\mathrm{Y}})^i \\
& + \ y'_{\mathrm{Pr}} y'_{\mathrm{Mg}} y'_{\mathrm{Y}} L^{\mathrm{MgM}}_{\mathrm{Pr,Mg,Y:Pr}} \Big] \\
& + y''_{\mathrm{Mg}} \Big[y'_{\mathrm{Pr}} y'_{\mathrm{Mg}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr,Mg:Mg}} (y'_{\mathrm{Mg}} - y'_{\mathrm{Pr}})^i \\
& + y'_{\mathrm{Mg}} y'_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Mg}} (y'_{\mathrm{Mg}} - y'_{\mathrm{Y}})^i \\
& + y'_{\mathrm{Pr}} y'_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr,Y:Mg}} (y'_{\mathrm{Pr}} - y'_{\mathrm{Y}})^i \\
& + \ y'_{\mathrm{Pr}} y'_{\mathrm{Mg}} y'_{\mathrm{Y}} L^{\mathrm{MgM}}_{\mathrm{Pr,Mg,Y:Mg}} \Big] \\
& + y''_{\mathrm{Y}} \Big[y'_{\mathrm{Pr}} y'_{\mathrm{Mg}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr,Mg:Y}} (y'_{\mathrm{Mg}} - y'_{\mathrm{Pr}})^i \\
& + y'_{\mathrm{Mg}} y'_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Y}} (y'_{\mathrm{Mg}} - y'_{\mathrm{Y}})^i \\
& + y'_{\mathrm{Pr}} y'_{\mathrm{Y}} \sum {}^{i} L^{\mathrm{MgM}}_{\mathrm{Pr,Y:Y}} (y'_{\mathrm{Pr}} - y'_{\mathrm{Y}})^i \\
& + \ y'_{\mathrm{Pr}} y'_{\mathrm{Mg}} y'_{\mathrm{Y}} L^{\mathrm{MgM}}_{\mathrm{Pr,Mg,Y:Y}} \Big]
\end{aligned} \tag{8}
$$

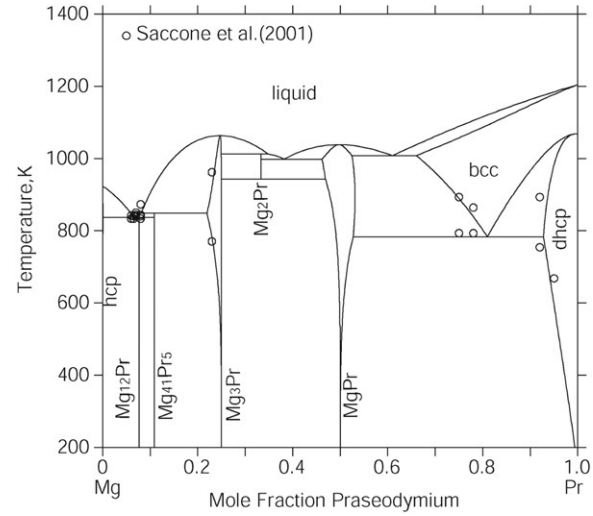

Fig. 2. Calculated Mg–Pr phase diagram by present thermodynamic description with experimental data measured by Saccone et al. [15].

region has a composition range 82–83 at.% Mg. In the present work, due to the small homogeneity range of Mg in the compound Mg_5M, it was treated as Mg_5(Pr, Y) and has a constant content of Mg atom. The Gibbs energy per mole of formula unit Mg_mPr_n (Mg_2Pr, $Mg_{12}Pr$, $Mg_{41}Pr_5$ and Mg_5M) is given by the following expression:

$$
\begin{aligned}
G_m^{\mathrm{Mg}_m\mathrm{Pr}_n} = {} & y_{\mathrm{Pr}} G^{\mathrm{Mg}_m\mathrm{Pr}_n}_{\mathrm{Mg:Pr}} + y_{\mathrm{Y}} G^{\mathrm{Mg}_m\mathrm{Pr}_n}_{\mathrm{Mg:Y}} + nRT(y_{\mathrm{Pr}} \ln y_{\mathrm{Pr}} \\
& + y_{\mathrm{Y}} \ln y_{\mathrm{Y}}) + y_{\mathrm{Pr}} y_{\mathrm{Y}} \sum_i {}^{i} L^{\mathrm{Mg}_m\mathrm{Pr}_n}_{\mathrm{Mg:Pr,Y}} (y_{\mathrm{Pr}} - y_{\mathrm{Y}})^i .
\end{aligned} \tag{9}
$$

The intermetallic compound Mg_3Pr in the Mg–Pr system has a homogeneity range, and is treated as Mg_3(Mg, Pr, Y). The Gibbs energy per mole of formula unit Mg_3Pr is expressed as follows:

$$
\begin{aligned}
G_m^{\mathrm{Mg}_3\mathrm{Pr}} = {} & y_{\mathrm{Mg}} G^{\mathrm{Mg}_3\mathrm{Pr}}_{Mg:\mathrm{Mg}} + y_{\mathrm{Pr}} G^{\mathrm{Mg}_3\mathrm{Pr}}_{\mathrm{Mg:Pr}} + y_{\mathrm{Y}} G^{\mathrm{Mg}_3\mathrm{Pr}}_{\mathrm{Mg:Y}} \\
& + RT(y_{\mathrm{Mg}} \ln y_{\mathrm{Mg}} + y_{\mathrm{Pr}} \ln y_{\mathrm{Pr}} + y_{\mathrm{Y}} \ln y_{\mathrm{Y}}) \\
& + y_{\mathrm{Mg}} y_{\mathrm{Pr}} \sum_i {}^{i} L^{\mathrm{Mg}_3\mathrm{Pr}}_{\mathrm{Mg:Mg,Pr}} \left(y_{\mathrm{Mg}} - y_{\mathrm{Pr}}\right)^i \\
& + y_{\mathrm{Pr}} y_{\mathrm{Y}} \sum_i {}^{i} L^{\mathrm{Mg}_3\mathrm{Pr}}_{\mathrm{Mg:Pr,Y}} (y_{\mathrm{Pr}} - y_{\mathrm{Y}})^i \\
& + y_{\mathrm{Mg}} y_{\mathrm{Y}} \sum_i {}^{i} L^{\mathrm{Mg}_3\mathrm{Pr}}_{\mathrm{Mg:Mg,Y}} (y_{\mathrm{Mg}} - y_{\mathrm{Y}})^i \\
& + y_{\mathrm{Mg}} y_{\mathrm{Pr}} y_{\mathrm{Y}} L^{\mathrm{Mg}_3\mathrm{Pr}}_{\mathrm{Mg:Mg,Pr,Y}} .
\end{aligned} \tag{10}
$$

4. Optimizing procedure

To obtain a thermodynamic description of a multi-component system, a thermodynamic description of each lower-order system is necessary. The Mg–Pr and Mg–Y systems have been optimized by Guo and Du [3], Guo and Du [2] and Fabrichnaya et al. [1], respectively. In order to assure the consistency of the Gibbs energy of pure elements among the three binary systems, the Mg–Pr and Pr–Y binary systems are re-optimized in the present work. In the Pr–Y binary system,

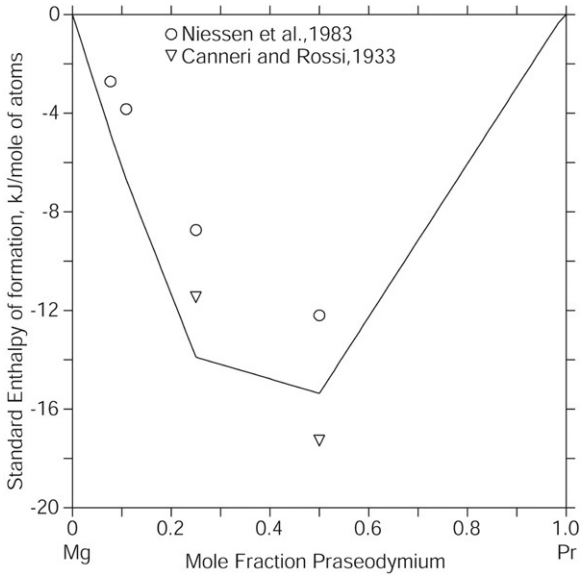

Fig. 3. Calculated standard enthalpy of formation at 298 K in the Mg–Pr system and comparison with the experimental data measured by Canneri and Rossi [17] and predicted values by the revised Miedema model of Niessen et al. [18].

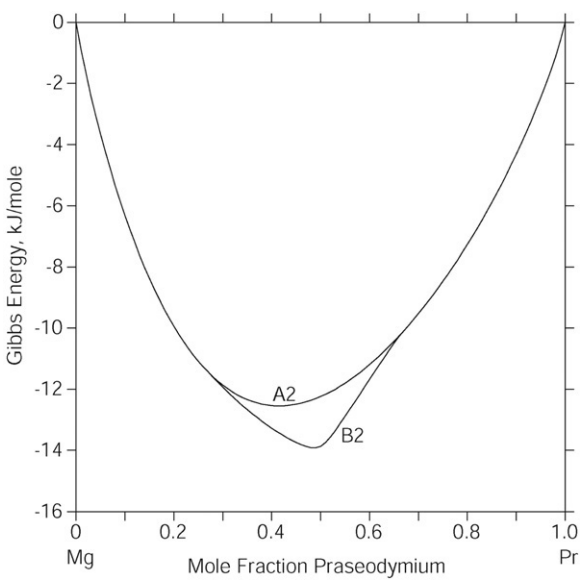

Fig. 4. Calculated Gibbs energies of the bcc solution (A2) and MgPr with CsCl-type structure (B2) at 900 K in the Mg–Pr system. The reference state is bcc for Mg and Pr.

the liquid and bcc phases are treated as the ideal solutions due to scarcity of experimental data. The thermodynamic parameters of Mg–Y binary system come from the result of Guo and Du [2].

The optimization was carried out by means of the optimization module PARROT of the thermodynamic software Thermo-Calc [28], which can handle various kinds of experimental data.

The thermodynamic model parameters for the Mg–Pr–Y system were optimized on the basis of the experimental information available in the literature [23]. The thermodynamic descriptions of the liquid phase as well as of the bcc, dhcp

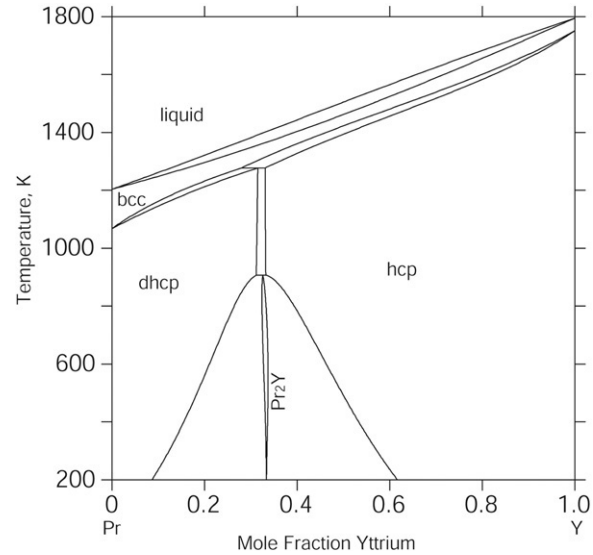

Fig. 5. Calculated Pr–Y phase diagram by the present thermodynamic description.

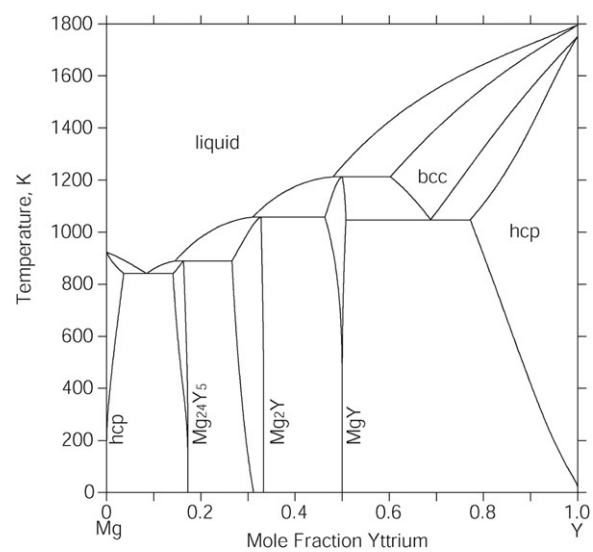

Fig. 6. Calculated phase diagram of the Mg–Y system optimized by Guo and Du [2].

and hcp phases were obtained by a combination of the corresponding Gibbs energy functions from the assessments of binary systems using Muggianu interpolation of binary excess terms [29]. The interaction parameter ${}^{i}L^{\phi}_{\mathrm{Mg,Pr}}$, ${}^{i}L^{\phi}_{\mathrm{Pr,Y}}$ and ${}^{i}L^{\phi}_{\mathrm{Mg,Y}}$ were taken from three binary systems. The ternary interaction parameters were fixed at zero as a result of lack of information regarding the solidification behaviors and the homogeneity ranges of ternary solution phases.

For the compounds in the ternary Mg–Pr–Y system, the parameters $G^{\mathrm{Mg_3Pr}}_{\mathrm{Mg:Pr}}$, $G^{\mathrm{Mg_2Pr}}_{\mathrm{Mg:Pr}}$, $G^{\mathrm{MgM}}_{\mathrm{Mg:Pr}}$, $G^{\mathrm{MgM}}_{\mathrm{Pr:Mg}}$, $G^{\mathrm{MgM}}_{\mathrm{Mg:Y}}$, $G^{\mathrm{MgM}}_{\mathrm{Y:Mg}}$, $G^{\mathrm{Mg_2Y}}_{\mathrm{Mg:Y}}$, $G^{\mathrm{Mg_2Y}}_{\mathrm{Y:Mg}}$, $G^{\mathrm{Mg_{24}Y_5}}_{\mathrm{Mg:Y:Y}}$, $G^{\mathrm{Mg_{24}Y_5}}_{\mathrm{Mg:Mg:Y}}$, $G^{\mathrm{Pr_2Y}}_{\mathrm{Pr:Y}}$, $G^{\mathrm{Pr_2Y}}_{\mathrm{Y:Pr}}$, ${}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg:Mg,Pr}}$, ${}^{1}L^{\mathrm{MgM}}_{\mathrm{Mg:Mg,Pr}}$, ${}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg,Pr:Pr}}$, ${}^{1}L^{\mathrm{MgM}}_{\mathrm{Mg,Pr:Pr}}$, ${}^{0}L^{\mathrm{MgM}}_{\mathrm{Y:Mg,Y}}$, ${}^{1}L^{\mathrm{MgM}}_{\mathrm{Y:Mg,Y}}$, ${}^{0}L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Y}}$, ${}^{1}L^{\mathrm{MgM}}_{\mathrm{Mg,Y:Y}}$, ${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Mg:Mg,Y}}$, ${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Y:Mg,Y}}$, ${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Mg,Y:Mg}}$, ${}^{0}L^{\mathrm{Mg_2Y}}_{\mathrm{Mg,Y:Y}}$, ${}^{0}L^{\mathrm{Pr_2Y}}_{\mathrm{Pr,Y:Pr}}$ and ${}^{0}L^{\mathrm{Pr_2Y}}_{\mathrm{Pr,Y:Y}}$ were taken from

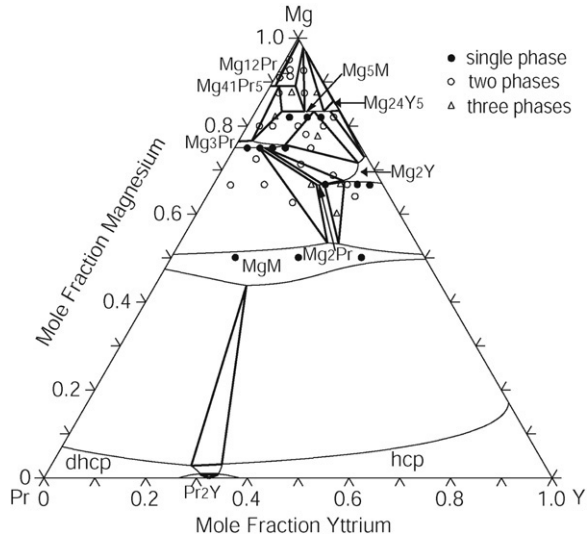

Fig. 7. Calculated isothermal section of the Mg–Pr–Y system at 773 K by the present thermodynamic description and comparison with experimental data determined by Giovannini et al. [23].

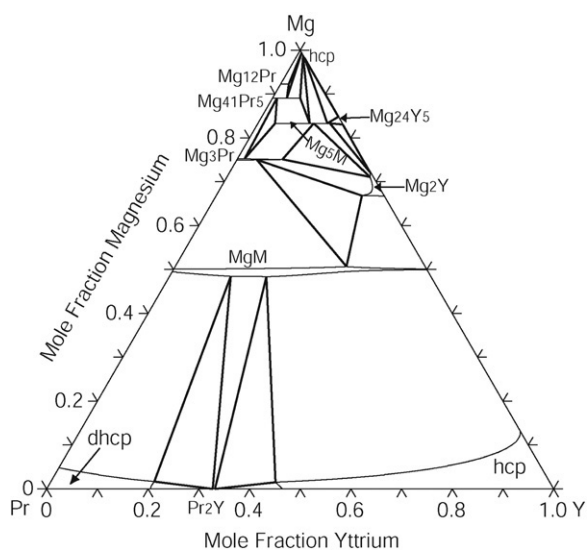

Fig. 8. Calculated isothermal section of the Mg–Pr–Y system at 573 K by the present thermodynamic description.

the thermodynamic descriptions of the corresponding binary systems. The parameters $G^{Mg_5M}_{Mg:Y}$, $G^{Mg_5M}_{Mg:Pr}$, ${}^0L^{Mg_5M}_{Mg:Pr,Y}$, $G^{Mg_3Pr}_{Mg:Y}$, $G^{Mg_2Pr}_{Mg:Y}$, ${}^0L^{Mg_2Pr}_{Mg:Pr,Y}$, ${}^1L^{Mg_2Pr}_{Mg:Pr,Y}$, $G^{Mg_2Y}_{Mg:Pr}$, $G^{Mg_{24}Y_5}_{Mg:Pr:Y}$ and ${}^0L^{Mg_{24}Y_5}_{Mg:Mg,Pr,Y:Y}$ were optimized according to the compounds in the ternary system.

5. Results and discussions

A thermodynamic description of the Mg–Pr–Y system obtained in the present work is shown in Table 1. The invariant equilibria in the Mg–Pr system and the Pr–Y system are listed in Tables 2 and 3, respectively. Satisfactory agreement is obtained between the calculations and experiments. The predicted invariant equilibria in the Mg–Pr–Y system are listed in Table 4.

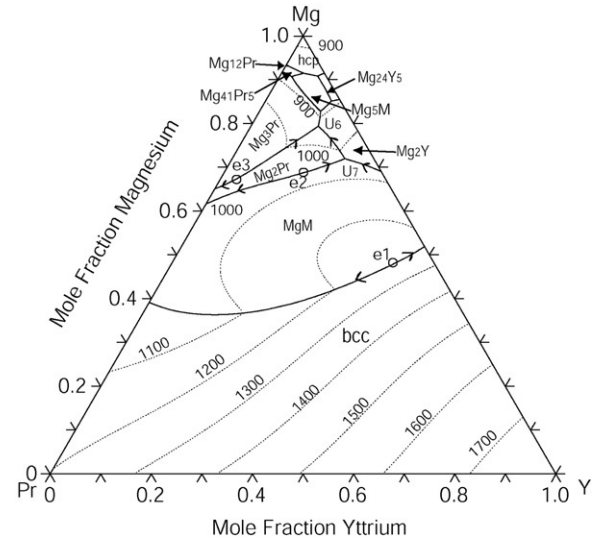

Fig. 9a. Calculated projection of the liquidus surfaces in the Mg–Pr–Y system by the present thermodynamic description.

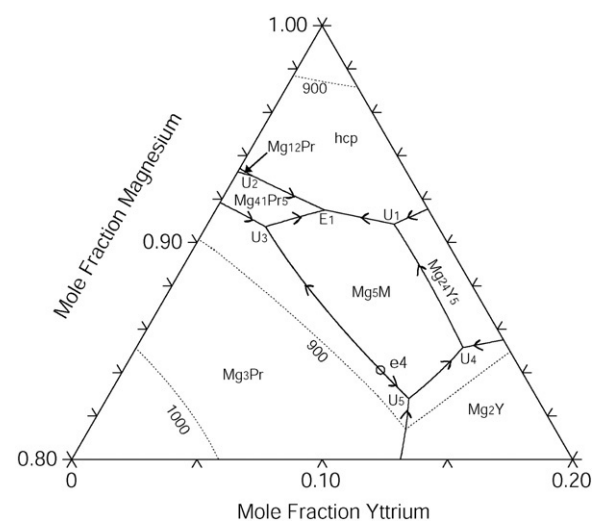

Fig. 9b. Enlarged section of predicted projection of the liquidus surfaces.

Fig. 2 represents the phase diagram of the Mg–Pr system calculated by present thermodynamic parameters with the experimental data [15]. The difference between the present result and the result of Guo and Du [3] is the Gibbs energy of the compound MgPr. In the present work, the compound MgPr with B2 structure was treated to be in phase with the disordered A2 phase. Due to the application of the Redlich–Kister equation to model the Gibbs energy of liquid phase, an unwanted miscibility gap occurs in the liquid above 4500 K. Kaptay [30] and Arroyave and Liu [31] pointed out that this expression of the Redlich–Kister equation may become problematic, especially in cases where the thermodynamic description is intended to be valid over a wide temperature range. Usually, the excess enthalpies and entropies of mixing have the same sign. Thus, there is always a temperature at which the ${}^iL^{\phi}_{Mg,Pr}$ parameter changes sign. When this change is from negative to positive, and if this effect is more dominant

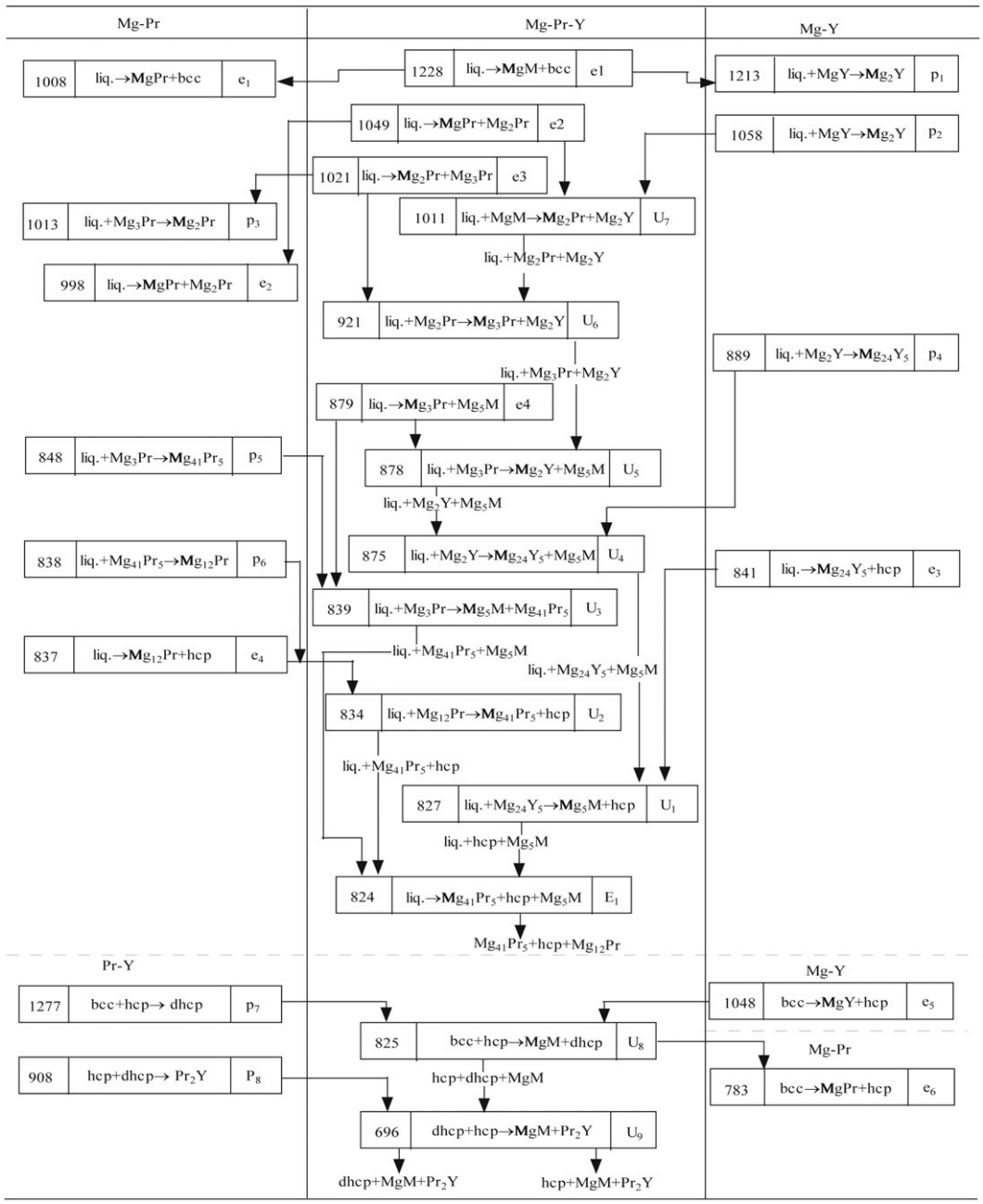

Fig. 10. Reaction scheme in the Mg–Pr–Y system.

than the $RT\sum_i x_i \ln x_i$ term due to ideal mixing,then spurious miscibility gaps tend to appear.

Fig. 3 shows the standard enthalpy of formation at 298 K in the Mg–Pr system and comparison with the experimental data measured by Canneri and Rossi [17] and the predicted values of the revised Miedema model [18]. Satisfactory agreement is obtained. Fig. 4 is the calculated molar Gibbs energies of the bcc solution (A2) and MgPr with CsCl-type structure (B2) at 900 K in the Mg–Pr system. It represents the coupling of the ordered bcc-B2 and the disordered bcc-A2 in the Mg–Pr system.

The Pr–Y phase diagram calculated by means of the thermodynamic parameters is presented in Fig. 5. In the present work, the unreasonable phenomenon that the solid solution phases dhcp(Pr) and hcp(Y) form a homogeneous solid solution in the assessed Pr–Y phase diagram (see Fig. 1) was modified according to the experimental phase diagram of La−Y system determined by Spedding et al. [32].

Fig. 6 is the Mg–Y phase diagram optimized by Guo and Du [2]. Figs. 7 and 8 are the isothermal sections of the Mg–Pr–Y system calculated at 773 K and 573 K respectively, by the present thermodynamic description. Satisfactory agreement is obtained between the calculations and experiments except for the single phase of compound Mg_5M. The difference between the calculated and experimental data for the compound Mg_5M is due to the fact that the content of Mg is treated to be constant in the present work. Figs. 9a and 9b is the calculated projection of the liquidus surfaces of the Mg–Pr–Y system according to the present thermodynamic description,

the open circles are the highest points in the single variable line. Fig. 10 shows the predicted entire reaction scheme for the Mg–Pr–Y system. The invariant reactions above the dashed line are relative to the liquid phase.

6. Conclusions

The phase relations and the thermodynamic description in the Mg–Pr–Y system were critically evaluated from the experimental information available in the literature. A set of self-consistent thermodynamic parameters describing the Gibbs energy of each individual phase as a function of composition and temperature was derived. The projection of the liquidus surfaces and the reaction scheme for the entire Mg–Pr–Y system have been predicted. With the thermodynamic description available, one can now make various calculations that are of practical interest.

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